

WADV SERIES (TIMER, ZERO LOSS & HIGH PRESSURE)

Industrial Automatic Condensate Drain Valves

Electronic Timer Drains, Capacitive Zero-Air-Loss Drains & 40-Bar High Pressure Drain Systems for Guaranteed Condensate Evacuation Without Energy Waste

**ZERO AIR LOSS**

Capacitive Sensor
Discharges Water Only

UP TO 40 BAR

Standard 16 Bar & High
Pressure PET Models

INTEGRATED MESH

Stainless Strainer
Prevents Orifice Clogging

TEST BUTTON

Instant Manual Override
& Diagnostic Cycle

THERMODYNAMIC ARCHITECTURE & ENGINEERING

Operational Overview & Engineering Advantages

Continuous industrial duty engineered for 45°C tropical ambient conditions

Win Equipments WADV Series automatic condensate drain valves eliminate moisture and oil slurry accumulated in compressor aftercoolers, air receivers, refrigerated dryers, and drop legs. Condensate buildup damages pneumatic cylinders and contaminates finished goods, while manual draining is frequently neglected. WADV valves provide fully automated, trouble-free drainage across three core technologies: electronic analog timer valves, intelligent capacitive zero-air-loss drains, and 40-bar high-pressure drain systems.

Working Principle: In WADV-T16 timer valves, a solid-state microcontroller energizes a direct-acting brass solenoid valve at preset intervals (0.5 to 10 seconds ON time; 0.5 to 45 minutes OFF time), discharging accumulated slurry through an integrated stainless steel mesh filter. In WADV-Z16 zero-air-loss valves, a contactless capacitive sensor monitors liquid level inside an aluminum collection chamber; as condensate reaches peak level, the pilot valve opens to exhaust 100% water before snapping shut before any compressed air escapes, saving thousands of rupees in compressed air energy annually.

SYSTEM INTEGRATION & P&ID FLOW ARCHITECTURE

Condensate Influx → Upstream Ball-Valve Strainer → Capacitive Sensing Chamber → Pilot Actuation Orifice → 100% Liquid Discharge (0 CFM Air Lost) → Oily Water Separator

Capacitive Zero Air Loss Technology (WADV-Z16)

Non-contact electronic capacitive sensor senses liquid level through the chamber wall with zero moving floats. Exhausts 100% liquid and closes before air can escape, saving up to ₹38,000 annually per valve.



Built-In Stainless Steel Mesh Strainer

WADV-T16 features a forged brass ball-valve strainer integrated directly into the valve body. Captures pipe scale, rust flakes, and heavy slurry before the solenoid orifice, eliminating stuck-open valve failures.

High-Pressure 40-Bar Heavy Duty Model (WADV-HP40)

Specially engineered with heavy-duty stainless steel valve seats and high-temperature coils rated for 40 bar g operating pressure, purpose-built for PET stretch blow molding booster compressors.



Manual Test Override & LED Status Indicators

Features dual bright LED indicators for power and valve-open status, alongside a tactile push-to-test button allowing maintenance operators to test drain functionality instantly during audits.

CERTIFIED ENGINEERING RATINGS

Technical Specifications Matrix

Full standard model range & operational parameters

Standard Design Conditions: Supply Voltage: 220V AC / 50Hz (Optional 24V DC / 110V AC) | Working Pressure: 0 to 16 bar g (HP model up to 40 bar g) | Ambient Temp: 1°C to 60°C | Enclosure: IP65 Heavy Duty | Body: Forged Brass / Stainless Steel

MODEL CODE	DRAIN TECHNOLOGY	OPERATING PRESSURE	IN/OUT CONN	DISCHARGE CYCLE	BODY METALLURGY	POWER SUPPLY
WADV-T16	Electronic Timer Drain	0–16 bar g	1/2" BSP In / Out	0.5–10s ON / 0.5–45m OFF	Forged Brass + SS Mesh	220V / 1Ph / 50Hz
WADV-Z16	Capacitive Zero-Air-Loss	0.8–16 bar g	1/2" BSP In / 1/4" Out	Demand-based (0 CFM Loss)	Die-Cast Aluminum	220V / 1Ph / 50Hz
WADV-HP40	High Pressure Timer	0–40 bar g	1/2" BSP High-Press	0.5–10s ON / 0.5–45m OFF	Heavy Stainless Steel	220V / 1Ph / 50Hz
WADV-Z40	Zero-Loss High-Pressure	1.2–40 bar g	1/2" BSP High-Press	Demand-based (0 CFM Loss)	Stainless Steel Body	220V / 1Ph / 50Hz

ENGINEERING SELECTION & OPERATIONAL NOTES

- Zero-air-loss models feature a potential-free alarm contact for integration into central SCADA / PLC control rooms.
- Timer drains include a built-in isolation ball valve for easy inline strainer cleaning without depressurizing the main air header.
- Explosion-proof ATEX certified coils available on request for hazardous chemical atmospheres.

FIELD DEPLOYMENT & COMPETITIVE ANALYSIS

Applications & Industrial Benchmarks

Demonstrated operational benefits across high-reliability manufacturing sectors

- Air Compressor Aftercooler Moisture Draining** Instantly purges heavy condensate slugs discharged from compressor aftercoolers before reaching receiver vessels.
- Air Receiver Bottom Sump Drainage** Eliminates corrosive water pooling in receiver vessel dished bottoms, preventing internal shell pitting and wall thinning.
- Refrigerated & Desiccant Dryer Separators** Discharges condensed water from 3-in-1 heat exchangers without wasting precious compressed air volume.
- Factory Distribution Pipe Drop Legs** Automatically drains condensate accumulated at line low-points throughout extensive factory ring-main networks.
- PET Bottle Blow Molding Air Booster Systems** Withstands severe 40-bar shock pulsations and ensures zero moisture enters high-pressure stretch blow molds.

Energy Economics: Zero-Air-Loss WADV-Z16 vs. Timed Solenoid Drains		Standard ISO 9001:2015 Quality
Engineering Parameter	Win Equipments Engineered Solution	Standard / Imported Benchmark
Air Loss	0 CFM compressed air lost during discharge	Wastes 3–5 CFM air on every timer blast
Annual Power Cost	₹0 wasted air power cost per year	Wastes ₹38,000 to ₹55,000 in electrical power per valve
Slog Resistance	Generous pilot orifice handles heavy oil-water emulsions	Tiny solenoid orifice clogs frequently with rust slurry
Reliability	Contactless capacitive sensing with no mechanical float arms	Mechanical float drains jam open or fail to seal

MANUFACTURING STANDARDS & FACTORY VERIFICATION

Factory Quality Assurance & Consultation

Calibrated testing regimes, global certifications, and direct engineering contact

✓ Factory Acceptance Testing (FAT)

- 100% Factory pressure hold test at 1.5× rated pressure (24 bar for standard; 60 bar for HP)
- Over 10,000 cycle rapid firing endurance test under full operating pneumatic pressure
- IP65 waterproof enclosure spray test and electrical insulation dielectric testing
- Capacitive water level detection sensitivity calibration test across varying emulsion mixtures

★ Certified Standards & Accreditations

- ISO 9001:2015 Quality Management System
- IAF (International Accreditation Forum) Accredited
- DAC (Dubai Accreditation Center) Recognized
- Pre-dispatch run-testing under tropical temperature simulation



Manufacturer Warranty Assurance: 24 Months replacement warranty on zero-air-loss drains; 12 months on timer drain valves. Direct field service engineering and genuine replacement spare parts stocked at Arasur Works, Coimbatore.

WIN EQUIPMENTS

SAVE WATER AND POWER • ISO 9001:2015 (IAF & DAC ACCREDITED)

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